

C.W

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(*) Use mathematical induction to show,

$$\sum_{j=0}^n ar^j = \frac{ar^{n+1} - a}{r-1} \text{ where } r \neq 1.$$

Solve:

$$\sum_{j=0}^n ar^j = \frac{ar^{n+1} - a}{r-1}$$

$$\Rightarrow a + ar + ar^2 + ar^3 + \dots + ar^n = \frac{ar^{n+1} - a}{r-1} \quad (i)$$

Basic Step:

Let assume, the eqⁿ (i) is true for $n=0$

then, $LS = a$

$$RS = \frac{ar^{0+1} - a}{r-1}$$

$$= \frac{ar - a}{r-1}$$

$$= \frac{a(r-1)}{(r-1)} = a$$

$$\therefore LS = RS$$

Now does not exist in (i) for $n=0$

Induction Step:

Assume, the eqⁿ (i) is true for $n = m$.

where m is an integer,

then,

$$a + ar + ar^2 + \dots + ar^m = \frac{ar^{m+1} - a}{r - 1} \quad \text{--- (ii)}$$

Again, Assume the consecutive member

(i) of m is $(m+1)$. Then,

$$a + ar + ar^2 + \dots + ar^{m+1} = \frac{ar^{m+2} - a}{r - 1} \quad \text{--- (i)}$$

Now, add (ar^{m+1}) to the both sides of

(ii), we get:

$$a + ar + ar^2 + \dots + ar^m + ar^{m+1} = \frac{ar^{m+1} - a}{r - 1} + ar^{m+1}$$

$$= \frac{ar^{m+1} - a + (r-1)ar^{m+1}}{r-1}$$

$$= \frac{ar^{m+1} - a + ar^{m+2} - ar^{m+1}}{r-1}$$

$$= \frac{ar^{m+2} - a}{r-1}$$

So, the eqⁿ (i) is true for each $m \in \mathbb{N}$.

(*) Use M.I. to prove that the sum of first n odd positive integers is n^2 .

Answer:

The sum of first n odd positive integers is
 $1 + 3 + 5 + 7 + \dots + n$

So,
 $1 + 3 + 5 + 7 + \dots + (2n-1) = n^2 \dots (i)$

Basic Step:

[Last Note - go - Example 1]
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$$\text{H.W: } 1^2 + 3^2 + 5^2 + 7^2 + \dots + (2n+1)^2 = \frac{n(2n-1)(2n+1)}{3}$$